RES POLYTECHNIC, BANGALORE-91

CIE Theory Test III

Program	Computer Science and Engineering		Semester- III
Course Name	System and Network Administration		Test III
Course Code	25CS34I	Duration- 90 min	Marks-50
Name of the Course Coordinator: Vinay Kumar			
Note: Answer any one full question from each section. Each full question carries equal marks.			

Q. No	Questions	CL	CO	Marks
	SECTION I			
1	**a)** List the steps to create a bootable pen drive and install Windows 10/11 on a computer. **(5 marks)** **b)** Meera's lab laptop suddenly shut down during a power cut. Now it does not start normally and keeps showing the "Automatic Repair" screen again and again. Explain how you will identify and solve this problem using Safe Mode, System Restore, System Reset and Image Recovery. When will you choose each option? **(10 marks)** **c)** You are the lab administrator. A new student user account named CSE_LAB1_02 has to be created and the folder D:\Records has to be protected. Write the command-line (Command Prompt/PowerShell) commands to add the account, modify it, delete it, and to encrypt and decrypt the folder. Explain each command briefly. **(10 marks)**	L2, L3	2	25
	OR			
2	**a)** Differentiate between the FAT32, NTFS and exFAT file systems. **(5 marks)** **b)** Several computers in your lab take a long time to start and become very slow after one hour of use. Explain how you will find the cause using Task Manager, Resource Monitor and Event Viewer, and how you can improve the performance (startup applications, services, Disk Cleanup, malware check, driver update, disk health). **(10 marks)** **c)** Anil bought a new 3 TB disk, but his computer shows only 2 TB. He also wants to divide it into three drives. Explain the difference between MBR and GPT, and how he can create, format and shrink/extend partitions using Disk Management. **(10 marks)**	L2, L3	2, 3	25

Q. No	Questions	CL	CO	Marks
	SECTION II			
3	**a)** What is a VPN? Explain any two types of VPN. **(5 marks)** **b)** Rahul can open a website when he types its IP address, but the same website does not open when he types its name. Identify the problem and explain how you will diagnose and solve it using ipconfig, ping, tracert and flushing the DNS cache. What is the role of the DNS server and the default gateway here? **(10 marks)** **c)** A small office wants its staff to share a project folder with other computers in the network, but outsiders must not access it. Staff also want to work from home safely. Explain how you will share the folder, configure the Windows Firewall and network profile, and set up secure remote access. **(10 marks)**	L2, L3	4	25
	OR			
4	**a)** What is a virtual machine (hypervisor)? Give any two benefits of installing a VM for testing a new operating system. **(5 marks)** **b)** Your college is replacing 30 old computers in a lab. Define e-waste with examples. Explain how you will dispose of these computers using the e-waste lifecycle (collection, sorting and recovery) and mention the environmental and health impacts of improper disposal. Name the government websites where the latest rules can be checked. **(10 marks)** **c)** You are asked to plan a new computer lab for 30 students. Prepare a lab setup plan covering hardware purchase, installation, electrical safety measures and cybersecurity protection. **(10 marks)**	L2, L3	4, 5	25

Note for the Course Coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.

Sign of the Course Coordinator Signature of the HOD Signature of the IQAC Chairman